

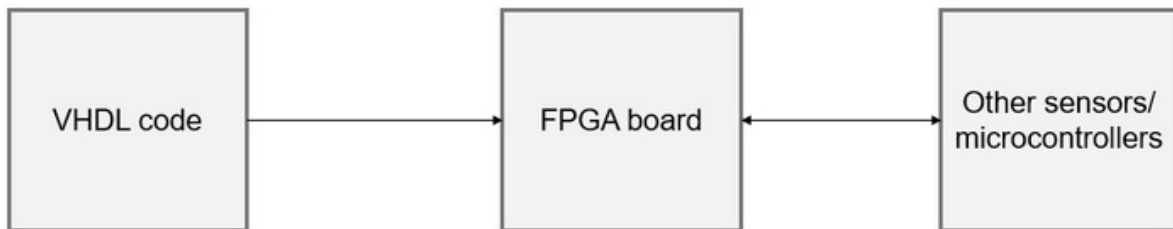
CSEN 605: Digital System Design

Winter 2025

Quartus Tutorial with DE10 lite board and MAX10 FPGA

In this tutorial, you will learn how to download a VHDL code on the DE10-lite FPGA board and use the various switches and LEDs to check the correctness of your circuit.

The overall diagram is shown below. Your VHDL codes are downloaded into the FPGA chip which then interacts with the on-board I/O such as switches, LEDs, and screens, or with out-of-board sensors and microcontrollers such as Arduino.



Part 1- Introduction

Before we start, let's check the DE10-lite board.

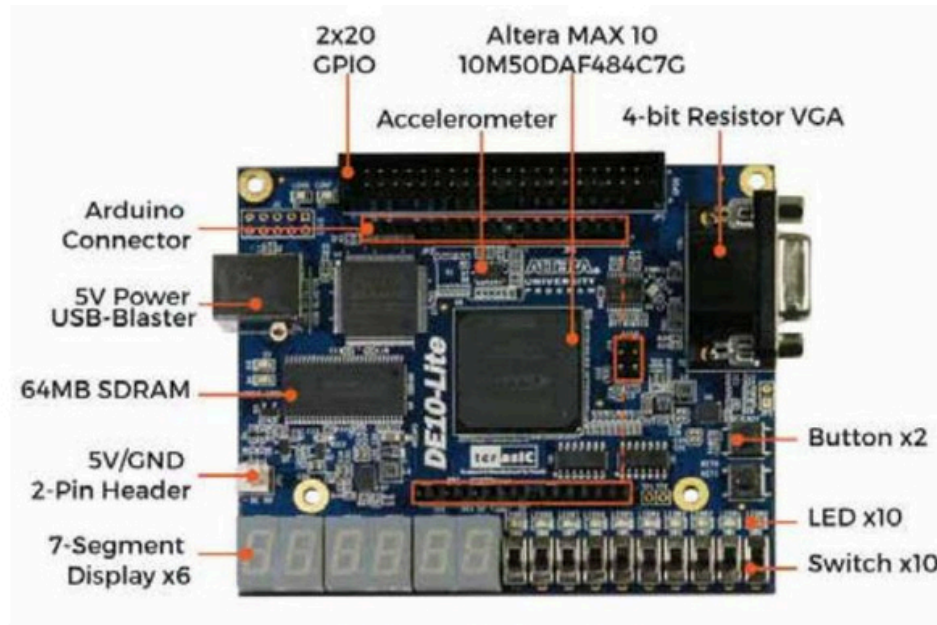


Figure 1-2 Development Board (top view)

The board contains:

- **FPGA Device:**
 - o Altera MAX 10 10M50DAF484C7G Device (might be another number)
 - o Integrated dual ADCs, each ADC supports 1 dedicated analog input and 8 dual function pins
 - o 50K programmable logic elements
 - o 1,638 Kbits M9K Memory
 - o 5,888 Kbits user flash memory
 - o 144 18 × 18 Multiplier
 - o 4 PLLs Programming and Configuration
- On-Board USB Blaster (Normal type B USB connector)
- **Memory Device**
 - o 64MB SDRAM, x16 bits data bus

- **Connectors**
 - 2x20 GPIO Header (General purpose I/O)
 - Arduino Uno R3 Connector, including six ADC channels.
- **Display**
 - 4-bit resistor-network DAC for VGA (With 15-pin high-density D-sub connector)
- **Switches, Buttons and LEDs**
 - 10 LEDs
 - 10 Slide Switches
 - 2 Push Buttons with De-bounced. (vibration of button is handled)
 - Six 7-Segments
- **Power**
 - 5V DC input from USB or external power connector.

Status LEDs

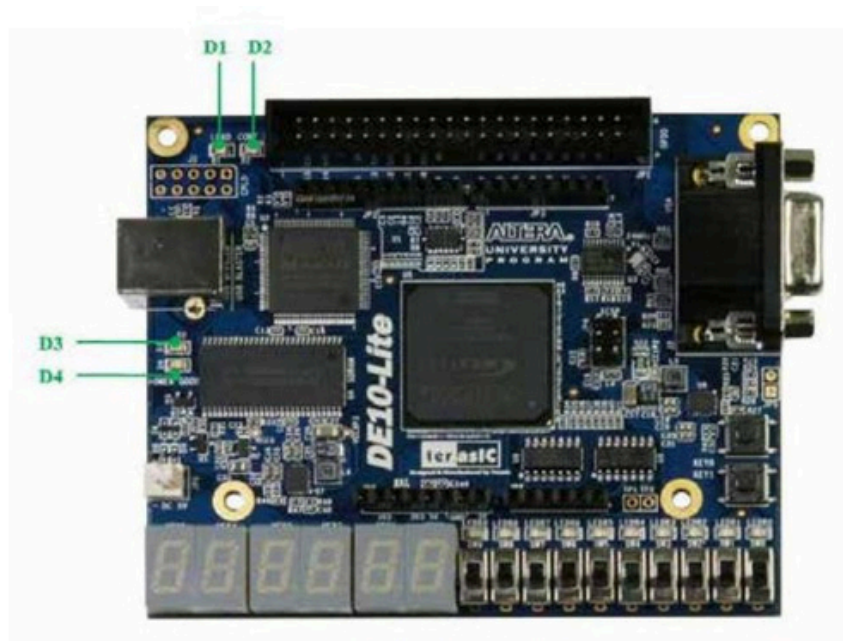


Figure 3-11 Status LED position

Table 3-1 Status LED

Reference	LED Name	Description
D1	ULED	Illuminates when the on-board USB-Blaster is working
D2	CONF_DONE	Illuminates when the FPGA is successfully configured.
D3	5V	Illuminates when Input power is active. Not Installed.
D4	Power Good	Illuminates when board power system is OK.

Switches and LEDs

Switches and LEDs (and all others I/O) have specific pins on the board so when you wish to connect your inputs/outputs in VHDL later you will use this pin assignment.



Figure 3-15 Connections between the slide switches and MAX 10 FPGA

Table 3-4 Pin Assignment of Slide Switches

Signal Name	FPGA Pin No.	Description	I/O Standard
SW0	PIN_C10	Slide Switch[0]	3.3-V LVTTTL
SW1	PIN_C11	Slide Switch[1]	3.3-V LVTTTL
SW2	PIN_D12	Slide Switch[2]	3.3-V LVTTTL
SW3	PIN_C12	Slide Switch[3]	3.3-V LVTTTL
SW4	PIN_A12	Slide Switch[4]	3.3-V LVTTTL
SW5	PIN_B12	Slide Switch[5]	3.3-V LVTTTL
SW6	PIN_A13	Slide Switch[6]	3.3-V LVTTTL
SW7	PIN_A14	Slide Switch[7]	3.3-V LVTTTL
SW8	PIN_B14	Slide Switch[8]	3.3-V LVTTTL
SW9	PIN_F15	Slide Switch[9]	3.3-V LVTTTL



Figure 3-16 Connections between the LEDs and MAX 10 FPGA

Table 3-5 Pin Assignment of LEDs

Signal Name	FPGA Pin No.	Description	I/O Standard
LED0	PIN_A8	LED [0]	3.3-V LVTTTL
LED1	PIN_A9	LED [1]	3.3-V LVTTTL
LED2	PIN_A10	LED [2]	3.3-V LVTTTL
LED3	PIN_B10	LED [3]	3.3-V LVTTTL
LED4	PIN_D13	LED [4]	3.3-V LVTTTL
LED5	PIN_C13	LED [5]	3.3-V LVTTTL
LED6	PIN_E14	LED [6]	3.3-V LVTTTL
LED7	PIN_D14	LED [7]	3.3-V LVTTTL
LED8	PIN_A11	LED [8]	3.3-V LVTTTL
LED9	PIN_B11	LED [9]	3.3-V LVTTTL

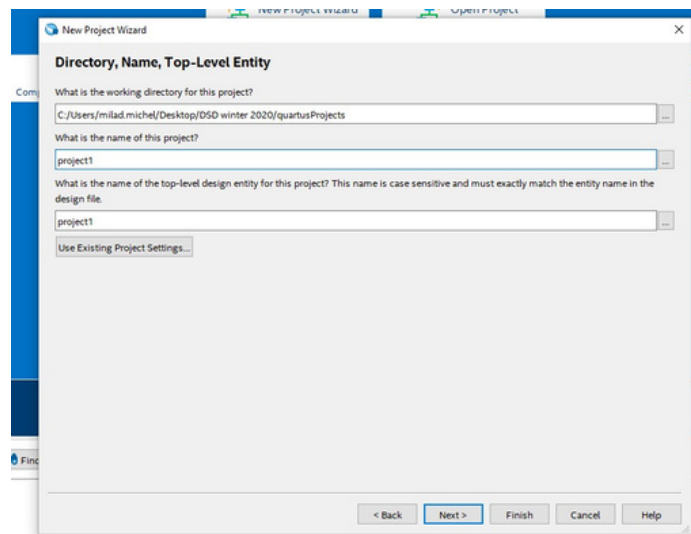
The full details can be found in the official datasheet of the de10-lite board.

Part 2- Setup and coding

In this part, you will learn how to implement the function $F = A'B' + C$ on the FPGA board and test its correctness using the switches and LEDs.

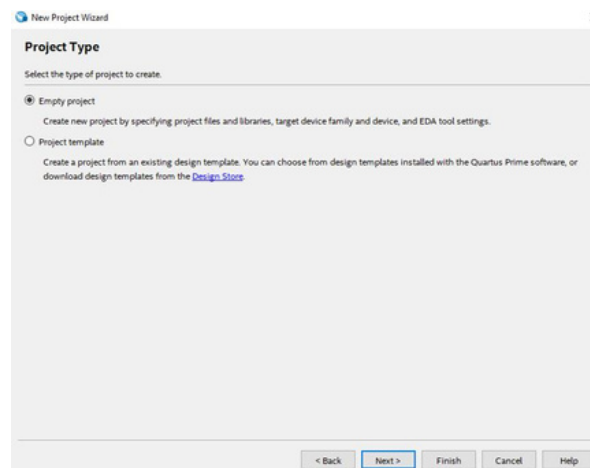
Step 1

- Open Quartus Lite and create a new project using the Wizard. Name it “project1” and choose a directory of your own choice. Do not use the default directory.
- Click next.



Step 2

- Choose **empty project** and click on **next**.



Step 3

- Choose the right board by choosing max 10 in the family field and then use the name filter field to choose the correct Altera FPGA chip on your board.
- **N.B. The name should be written on the Altera chip itself and is usually 10M50DAF484C7G or 10M50DAF484C6GES. (or something else if it's not max10)**

Family, Device & Board Settings

Device Board

Select the family and device you want to target for compilation.
You can install additional device support with the Install Devices command on the Tools menu.

To determine the version of the Quartus Prime software in which your target device is supported, refer to the [Device Support List](#) webpage.

Device family

Family: MAX 10 (DA/DF/DC/SA/SC)

Device: All

Target device

☐ Auto device selected by the Fitter

☒ Specific device selected in 'Available devices' list

☐ Other: n/a

Show in 'Available devices' list

Package: Any

Pin count: Any

Core speed grade: Any

Name filter: 10m50daf484c

☒ Show advanced devices

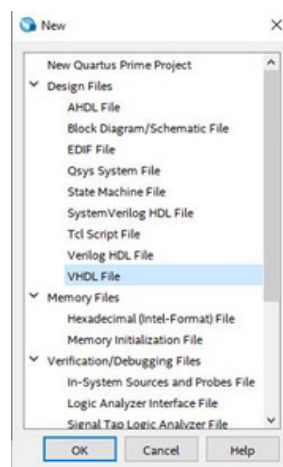
Available devices:

Name	Core Voltage	LEs	Total I/Os	GPIOs	Memory Bits	Embedded multiplier 9-bit elements	PLLs	Global Clocks	Maximum User Flash Memo
10M50DAF484C6GES	1.2V	49760	360	360	1677312	288	4	20	11534336
10M50DAF484C7G	1.2V	49760	360	360	1677312	288	4	20	11534336
10M50DAF484C8G	1.2V	49760	360	360	1677312	288	4	20	11534336
10M50DAF484C8GES	1.2V	49760	360	360	1677312	288	4	20	11534336

Once you find it, click on it and then click on finish.

Step 4

- Create the VHDL file using file->new-> VHDL file



Then, write the following code:

```
library ieee;
use ieee.std_logic_1164.all;

entity project1 is
port( A,B,C: in std_logic; F: out std_logic);
end project1;

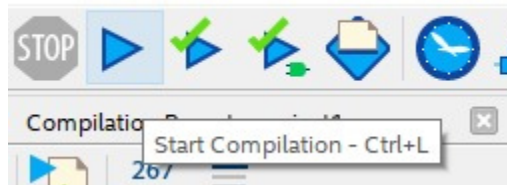
architecture arch of project1 is
begin
F<= (NOT A and NOT B) OR C;
end arch;
```

Step 5

- Save this file and name it project1 (same as your entity name)
- N.B. Make sure you place it in the same folder as your project. (next to **db** folder, not inside it)

Step 6

- Compile your code using **Processing-> start compilation** or by clicking on the blue arrow.



Then wait until Quartus compiles your code, you should get no errors. Warnings are fine at the point.

Tasks			Compilation			
Task						
✓	▼	▶	Compile Design			00:00
✓	>	▶	Analysis & Synthesis			00:00
✓	>	▶	Fitter (Place & Route)			00:00
✓	>	▶	Assembler (Generate programming files)			00:00
✓	>	▶	Timing Analysis			00:00
	>	▶	EDA Netlist Writer			
		■	Edit Settings			
		🔧	Program Device (Open Programmer)			

Part 3- Download to FPGA board

Usually, there's a step before downloading this code to FPGA and it's functional simulation using modelsim which you have seen in a previous tutorial to make sure of the functional correctness.

Step 3.1 – Pin assignment

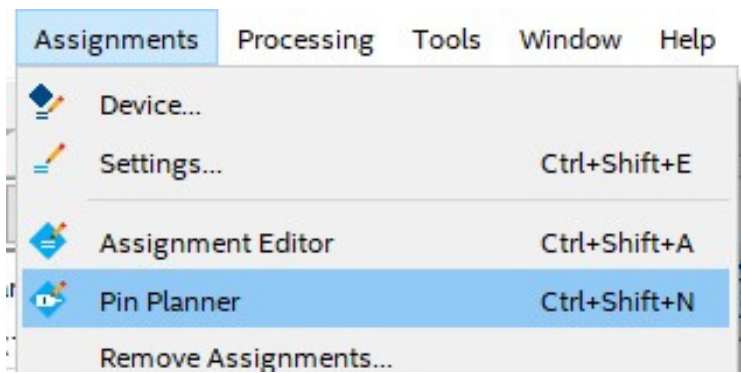
Downloading your code into the FPGA board should follow several steps, starting with pin assignment.

We need to assign each input and output in our code to a specific pin on the board.

In this tutorial, we will assign inputs A, B and C to switches 0, 1 and 2, respectively. Those 3 switches are actually pins C10, C11 and D12 (check table in page 4)

We will also connect the output F to LED0 which is pin A8.

- To do this, go to **assignments -> pin planner**



- A new window appears:

Top View - Wire Bond
MAX 10 - 10M50DAF484C7G

Node Name	Direction	Location	I/O Bank	VREF Group	Fitter Location	I/O Standard	Reserved	Current Strength	Slew Rate	Differential Pair	Strict Preservation
A	Input	PIN_C10	7	B7_NO	PIN_C10	2.5 V		12mA (default)			
B	Input	PIN_C11	7	B7_NO	PIN_C11	2.5 V		12mA (default)			
C	Input	PIN_D12	7	B7_NO	PIN_D12	2.5 V		12mA (default)			
F	Output	PIN_A8	7	B7_NO	PIN_A8	2.5 V		12mA (default)	2 (default)		

To assign inputs and outputs, you can **drag and drop** from the I/O list below into the desired pin on the board.

- For example, click on A and drag it to pin C-10.
- Do the same for B to pin C-11 and C to pin D-12.
- Drag and drop F to pin A-8.

Step 3.2

Recompile your project.

Step 3.3- Load the program onto the board

Here we might face a problem if it's your first time doing so: the computer can't understand your USB Connection. So To solve this follow these steps, if this is not your first time, skip to the last point.

1. Plug the board via USB-Blaster into your PC.
2. Open the Device and Printers (Control Panel | Devices and Printers).
3. Under Unspecified, USB Blaster should be listed. Right mouse click on this and then select Properties.
4. Select the Hardware tab and select Properties.
5. A new window should pop up with the General tab already selected. Select Change Settings.
6. Again a new window should pop up with the General tab already selected. Select Update.
7. Select Browse my computer for driver software.
8. Find \quartus\drivers\
 - (Note 1: Your altera file is located at the location you selected when you first installed quartus. The location listed in this document is the default location) ▪

(Note 2: **Stop** at the drivers folder, i.e., do NOT go deeper by opening a folder within the drivers folder)

1. Select OK. Make sure the proper path was selected then select Next.
2. If the Windows security window pops up Check the Always trust software from "Altera Corporation" box and select Install.

Step 3.4- Run the FPGA board

1. When you've finally connected your FPGA to your PC, Go to Tools ⇒ Programmer (or you know, find the shortcut button next to compile)
2. Click on **start**
3. Try a different combination of switches to monitor the output LED and check the correctness of your code.

